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AI/ML Intern

OVERALL PROJECT RESEARCH

Introduction

After analyzing the workflow diagrams and system architecture provided for the Raritone platform, I gained a comprehensive understanding of how the entire ecosystem functions. Initially, I viewed Raritone primarily as an AI Virtual Try-On application. However, after studying the workflows, I understood that it is actually a complete AI-powered fashion commerce platform that connects vendors, customers, backend services, analytics systems, and artificial intelligence modules into a unified solution.

This research helped me understand not only the responsibilities of the AIML team but also how the Backend and Data Analytics teams contribute to the overall system.

The Backend Team's Role

The backend team serves as the core foundation of the platform. Their responsibility is to ensure smooth communication between all modules and maintain the overall business logic of the application.

From the workflow, I observed that backend services are responsible for user authentication, product management, inventory tracking, order processing, logistics integration, payment handling, vendor verification, and cloud storage management.

Without the backend layer, the AI systems would not receive the required data, and the analytics team would not have access to meaningful business information. This made me understand that the backend team acts as the bridge connecting vendors, customers, AI services, and databases.

The technology and tools used by the backend team includes:

1. Java - Backend application development
2. REST APIs - Communication between frontend, backend, and AI services
3. PostgreSQL - Database management
4. MySQL - Relational data storage
5. AWS S3 - Product image and media storage
6. Firebase Authentication - User authentication and login management
7. JWT - Secure API authorization
8. Docker - Application containerization
9. AWS Cloud Services - Deployment and infrastructure management

Data Analytics Team's Role

The analytics team focuses on transforming raw business data into actionable insights. Based on the workflow, I observed that the platform collects information related to customer behavior, product performance, sales trends, inventory levels, marketing campaigns, reviews, returns, and vendor performance.

The analytics team helps identify which products are performing well, which marketing campaigns generate better conversions, which customers are more likely to purchase again, and where operational improvements can be made.

I also noticed that many of the advanced AI features depend heavily on historical data collected and organized through analytics processes.

The tools and technologies used by Data analytics team are:

- Pandas - Data manipulation and cleaning
- NumPy - Numerical computations
- Power BI - Dashboard creation and reporting
- Tableau - Data visualization
- Excel - Data organization and reporting
- SQL - Querying business data
- Matplotlib - Data visualization and chart generation
- Seaborn - Statistical data analysis

AIML Team's Role (My team)

As a member of the AIML team, I focused on understanding how artificial intelligence enhances the overall Raritone experience. The workflow shows that AI is integrated into multiple stages of the platform rather than being limited to virtual try-on functionality.

One of the primary AI responsibilities is enabling realistic virtual try-on experiences. This requires identifying human body landmarks, understanding user pose, segmenting garments, estimating body measurements, and accurately placing clothing onto user images.

I also observed that AI contributes to size recommendation systems by helping users select appropriate sizes based on body characteristics and product specifications. This can reduce return rates and improve customer satisfaction.

Another important observation was the presence of recommendation systems. AI can analyze user preferences, browsing patterns, and purchase history to recommend outfits and products that are most relevant to individual customers.

The workflow also highlights AI-driven business intelligence features such as demand forecasting, conversion prediction, trend analysis, and return prediction. These systems can help vendors make informed decisions regarding inventory planning, pricing strategies, and product launches.

Additionally, the AI Assistant for Vendors demonstrates how generative AI can simplify business operations by automatically generating product descriptions, tags, SEO content, and other supporting information.

Through this analysis, I realized that the AIML team's role extends far beyond virtual try-on technology. The team contributes to customer experience, operational efficiency, personalization, forecasting, and intelligent decision-making across the entire platform.

The technologies and tools used by my AI/ML team are:

- MediaPipe - Body Landmark Detection
- OpenPose - Human Pose Estimation
- OpenCV - Image Processing
- PyTorch - Deep Learning Models
- TensorFlow - AI Development
- VITON / HR-VITON - Virtual Try-On Systems
- Hugging Face - Pretrained AI Models
- Diffusion Models - Image Generation and Try-On Enhancement

Key Requirements Identified for the AIML Team

Based on my understanding of the workflow, the AIML team would require:

- Product image datasets
- Human pose and landmark datasets
- Garment segmentation datasets
- Fashion recommendation datasets
- Customer behavior data
- Sales and inventory data
- Return and refund history
- Product metadata and attributes
- Cloud-based model deployment infrastructure
- Integration with backend APIs for real-time predictions

These requirements are necessary to support the AI features shown throughout the workflow.

Understanding the Overall Project

The platform begins with vendors uploading products along with images, videos, and virtual try-on assets. These assets are stored in cloud storage while product information is maintained within the database. After verification and approval, the products become available to customers through APIs.

The customer-facing application allows users to browse products, view recommendations, use virtual try-on features, place orders, track deliveries, and provide reviews. Behind the scenes, multiple systems continuously interact to manage inventory, payments, logistics, analytics, and AI-powered features.

From the workflow, I understood that every action performed by a vendor or customer generates data that is processed by backend systems and later analyzed to improve business decisions and user experiences.

The tools and tech used by multiple and all teams in common:

TECHNOLOGY	TEAMS USING IT	PURPOSE
<u>Python</u>	AIML, Data Analytics	Machine Learning, Data Processing, Analytics, and Forecasting
<u>SQL / PostgreSQL</u>	Backend, Data Analytics	Storing, Managing, and Analyzing Platform Data
<u>AWS Cloud Services</u>	AIML, Backend	Hosting, Storage, and Model Deployment
<u>REST APIs</u>	AIML, Backend	Communication Between Applications, Databases, and AI Models
<u>Git & GitHub</u>	AIML, Backend, Data Analytics	Version Control and Team Collaboration

Conclusion

After studying the complete workflow, I understood that Raritone is a large-scale AI-powered fashion commerce platform rather than a standalone virtual try-on application. The success of the platform depends on close collaboration between the Backend, Data Analytics, and AIML teams.

The backend team provides the infrastructure and business logic, the analytics team generates insights from collected data, and the AIML team builds intelligent systems that enhance both customer experience and business performance.

This analysis helped me develop a clearer understanding of the overall architecture, the interaction between different teams, and the role artificial intelligence plays throughout the platform.